

AT POTTERS LODGE IN DEDZA

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**Centre for Epidemiological & Mathematical Research (CEMAR)
Private Bag 360
Chichiri- Blantyre 3
MALAWI**

CEMAR STAKEHOLDERS' ENGAGEMENT WORKSHOP AT POTTERS LODGE IN DEDZA DISTRICT

Welcome and Opening Remarks

The welcome remarks were delivered by Mr. Nicholas Adam, the Director of the Centre for Epidemiological and Mathematical Research (CEMAR). In his address, Mr. Adam introduced the purpose of the meeting, which was to formally present CEMAR to a broad group of stakeholders and to outline the organization's strategic plans to tackle health challenges related to infectious diseases, particularly malaria and HIV—through the use of mathematical modeling and data-driven approaches.

He emphasized that the meeting had two primary aims. First, it sought to facilitate the formulation of key policy questions that, once addressed, would provide evidence-based guidance for decision-making around malaria control in urban settings. Second, the meeting aimed to develop a proposed national framework for managing malaria in urban areas, recognizing the unique epidemiological and structural challenges associated with urban environments.

To achieve these objectives, the meeting was structured to first allow CEMAR to present its understanding of the issues at hand, articulate its mission, and share a broad overview of its technical capacity and ongoing work. This would set the stage for an informed and productive dialogue with stakeholders. Following CEMAR's presentation, the National Malaria Control Programme (NMCP) was presented a comprehensive presentation on the current malaria situation in Malawi, including trends, challenges, and ongoing interventions. This would provide critical context for discussions around urban malaria control and help identify opportunities for synergy between research and policy implementation.

Objectives and Goals of the Workshop

The primary goals of the workshop were:

1. **Introduction of CEMAR to key stakeholders:** Introduce the center, its mission, and strategic vision in the context of malaria and infectious disease modeling.
2. **Policy formulation:** Facilitate the development of critical policy questions related to malaria control.
3. **Urban malaria management framework:** Establish a national framework for malaria management in urban areas.
4. **Strengthening collaboration:** Enhance collaboration among academic institutions, government bodies, and research organizations to address malaria control issue

Introduction of the Participants

The Centre for Epidemiological and Mathematical Research (CEMAR) in Malawi, with funding support from the Applied Malaria Modelling Network (AMMNet), organized a one-day stakeholders' meeting on 9th April 2025 at Potter's Lodge in Dedza District, from 8:30 a.m. to 4:00 p.m. The primary purpose of the meeting was to engage key stakeholders in discussions around the use of mathematical modeling to inform decision-making in malaria control and other infectious disease challenges in Malawi.

The meeting brought together participants from a diverse range of institutions, including the National Malaria Control Programme (NMCP), academic and research institutions, and the CEMAR technical team. The participants represented expertise in epidemiology and public health, mathematical modeling, medicine, biostatistics, and public health policy.

Participants were drawn from the following organizations and institutions:

1. CEMAR Technical Team – comprising experts in modeling, epidemiology, and data science.

2. National Malaria Control Program (NMCP) – represented by Dr. Lumbani Munthali, the National Malaria Program Manager, along with other key officials from the Department of Malaria Prevention, Diagnostics, and Curative Services within the Ministry of Health.
3. Kamuzu University of Health Sciences (KUHeS) – represented by a team led by Professor Adamson Muula, with expertise in public health and infectious disease research.
4. Malawi University of Science and Technology (MUST) – represented by Dr. Peter Mwantobe, leading the delegation from the Department of Biostatistics and Mathematical Modeling.
5. University of Malawi (UNIMA) – represented by a team that joined virtually, led by Associate Professor Elias Mwakilama.

The full list of attendees, including their respective affiliations and contact details, is provided in the appendix of this report.

Presentation of the CEMAR

Nicholas Adam presented about CEMAR and people commented on the membership of CEMAR. He articulated the objectives and vision of CEMAR as detailed below:

Malawi continues to experience rapid outbreaks of infectious diseases; however, there has been limited focus on the use of modeling tools to support timely decision-making in disease control. Challenges such as inadequate data systems, weak infrastructure, limited technical skills, and a lack of appropriate tools and platforms have constrained the application of mathematical modeling in public health response. To address these gaps, the Centre for Epidemiological and Mathematical Research (CEMAR) was established to foster a data-driven approach in tackling public health threats.

CEMAR envisions becoming a regionally recognized research center that delivers impactful research and programs aimed at building a sustainable

future. The center focuses on generating knowledge relevant to real-world problems, with an emphasis on translational health research, environmental sustainability, and mathematical modeling for public safety. Its core objectives include improving surveillance systems, developing early warning tools for disease outbreaks, enhancing research collaborations, and supporting policy through evidence-based modeling. CEMAR's mission aligns with the Ministry of Health's goal of using innovation and research to strengthen health systems and protect population health.

Presentation from NMCP

The National Malaria Control Programme (NMCP) presented the current epidemiological situation of malaria in Malawi, highlighting a worrying upward trend in both malaria cases and deaths over the past three years. According to DHIS2 data, the number of confirmed malaria cases increased from 4.6 million in 2022 to 6.5 million in 2023, and further surged to 9.5 million in 2024. Deaths due to malaria also rose, with 1,829 reported in 2022, 1,670 in 2023, and 2,252 in 2024. The malaria incidence rate jumped to 467 cases per 1,000 population in 2024—an alarming 112% increase from the 2022 rate of 220 per 1,000.

The burden of malaria on the health system is also significant. In 2024, malaria accounted for 32% of all outpatient visits and 16% of all hospital admissions, up from 25% and 10% respectively in 2023. Malaria-related deaths made up 16% of total deaths in 2024, with a mortality rate of 11 per 100,000 population—up from 8 and 9 per 100,000 in 2023 and 2022, respectively. This growing burden underscores malaria's persistent threat to public health and its profound socioeconomic impact on families and the nation at large, making it a critical focus for public health interventions and resource allocation (more details in the appendix).

Some of the critical discussion and questions were central around:

1. Mass net distribution in schools

2. Social behavior change campaigns for the people who receive nets
3. The initiative to bring in genetically modified mosquitoes
4. Combined different preventive intervention (insecticide-treated bednets-ITNs and indoor residual insecticide spray-IRS) for better effectiveness
5. Encourage commercial entities to get involved in ITN businesses
6. Allow individuals to be eligible for IDS
7. Improve clinical practice in malaria management

Stakeholders Feedback and Discussions

The NMCP presented the following points of discussion:

- Malaria prevalence by microscopy is about three times higher in rural areas (12%) than in urban areas (4%).
- The prevalence of malaria in children aged 6-59 months by microscopy increases with decreasing wealth quintile, from 4% among those in the highest wealth quintile to 15% among those in the lowest wealth quintile.

Spatial and Temporal Dynamics:

- Identify specific areas eg, Informal settlements, areas near water bodies with higher malaria risk
- How do these patterns change with seasons or during different periods of the year?
- How does urbanization and infrastructure development influence malaria transmission?

Does increased population density or changes in housing types correlate with higher malaria risk?

How do changes in water management practices affect mosquito breeding sites?

Intervention Impact and Effectiveness:

- How effective are different malaria control interventions (e.g. insecticide treated nets, indoor residual spraying) in reducing malaria incidence?
- Can we quantify the impact of interventions on malaria prevalence and incidence?
- How does the effectiveness of interventions vary across different areas?
- What are the optimal strategies for allocating resources for malaria control?
- Which interventions provide the greatest return on investment?
- How can we tailor interventions to specific areas with different malaria risk profiles?

Predictive Modeling and Scenario Planning:

- Can we develop a predictive model to forecast malaria incidence under different scenarios (e.g. climate change, urbanization, intervention-scaling up)?
- What are the potential future trends in malaria incidence?
- How can we use these models to inform decision making for malaria control?
- How does the interaction between human behaviour, vector ecology, and parasite dynamics influence malaria transmission?
- What is the role of environmental factors e.g. climate, rainfall, temperature) in malaria transmission?

The academia from three universities said that most of the issues noted are doable and they will have the students do the areas that they have noted. The team from CEMAR also was in the positive view of having noted that the issues raised can be addressed. Participants made observations on 1) data sources,

2) the modelling interval, 3) Work done by the NMCP on some issues, and 4) plans that the NMCP has in relation to the issues besides the modelling need.

Participants deliberated on these key points and how they can be integrated into the modeling world in order to respond to these specific issues. On a similar note, members initiated discussions to build censuses on a framework for setting up a national network on integrated malaria approaches by different stakeholders. Participants agreed to come up with a national framework for malaria management in urban settings in Malawi. Based on WHO technical report of 2022 where the following were isolated and discussed as key points for the framework: leadership, strategic elements, response pillars and enabling elements.

Key Takeaways and Action Points

From the discussion, several key action points emerged

1. **Collaboration:** Establish a joint task force between CEMAR, academic and research institutions, and the National Malaria Control Programme (NMCP) to address data gaps and refine malaria modeling.
2. **Data Collection:** Systematically address issues with data sources and modeling intervals.
3. **Research Projects:** Academic institutions to assign student research projects related to the identified priority areas.
4. **Urban Framework:** Develop a national framework for malaria management in urban settings.
5. **Predictive Modeling:** Move forward with the development of predictive models for malaria incidence, considering environmental and intervention factors.

Way Forward

Building on the positive reception from academia and the modeling community, the next step involves concretizing the collaboration among universities and

research institutions, CEMAR, and the National Malaria Control Programme (NMCP) to ensure that the key areas identified—such as data sources, modeling intervals, and NMCP’s existing/planned interventions—are systematically addressed. Academic institutions will take the lead in assigning student research projects aligned with these priority areas, while CEMAR will provide technical guidance to ensure the modeling outputs are robust, policy-relevant, and actionable. A joint task team will be formed to coordinate efforts, harmonize methodologies, and establish data-sharing protocols with NMCP and other stakeholders.

Simultaneously, members endorsed the development of a national framework for malaria management in urban settings, anchored on WHO’s 2022 technical guidance. The framework will focus on four pillars: leadership and governance, strategic planning, operational response, and enabling systems such as financing, monitoring, and research. A consultative process will be initiated to draft the framework, bringing together representatives from health institutions, academia, local governments, and development partners. This integrated approach will not only enhance the role of mathematical modeling in decision-making but also support a unified, evidence-based response to urban malaria in Malawi.

Closing Remarks

The Director of the Centre for Epidemiological and Mathematical Research (CEMAR) expressed sincere gratitude to all participants for their active engagement and valuable contributions throughout the meeting. He acknowledged the collaborative spirit demonstrated by the National Malaria Control Programme (NMCP), academic institutions, and other partners, which enriched the discussions and helped shape a shared vision for malaria control in Malawi. He emphasized that the meeting marked an important step toward leveraging mathematical modeling and interdisciplinary collaboration to support data-driven decision-making in public health.

The National Malaria Control Programme (NMCP) Director reiterated CEMAR's commitment to continue working with all stakeholders to operationalize the proposed national framework and strengthen capacity for evidence-based malaria interventions. He closed the meeting by encouraging continued dialogue and partnerships beyond the meeting room to ensure lasting impact in the fight against malaria.

Appendix

Presentations

- ☐ CEMAR Stakeholders Meeting Presentation**
- ☐ Overview of Malaria Situation in Malawi for the Past 7 Years 31012025**
- ☐ Copy of Ruth Presentation**

Attendance List


CEMAR
Modeling for Improved Health

UNIVERSITY OF MALAWI
CRAFS /CEMAR-AMMnet STAKEHOLDER'S WORKSHOP



VENUE: POTTER LODGE, DEDZA

PROJECT NAME: ACE II AF CRAFS PROJECT

ATTENDANCE SHEET FORM

	NAME	INSTITUTION	DAYS	SIGNATURE	DATE
1	ANDREW MBEWE	CEMAR	1	<i>[Signature]</i>	9/04/25
2	Wongani Nyondo	CEMAR	1	<i>[Signature]</i>	09/04/25
3	MURRAY KUNGULINAM	NMCP	1	<i>[Signature]</i>	9/4/25
4	YOLAN KAMEME	CEMAR	1	<i>[Signature]</i>	9/4/25
5	Hopson Gausi	CEMAR	1	<i>[Signature]</i>	09/04/25
6	Peter Mwanuzi	NMCP	1	<i>[Signature]</i>	09/04/25
7	Precious Chingoma	NMCP	1	<i>[Signature]</i>	09/04/25
8	Al-jannat Sadat	NMCP	1	<i>[Signature]</i>	09/04/25
9	Mpende Chisanga	NMCP	1	<i>[Signature]</i>	9/4/25
10	Victoria Mirenda	NMCP	1	<i>[Signature]</i>	9/4/25
11	Akuzike Tauri	NMCP	1	<i>[Signature]</i>	9/4/25
12	Dr LUMBANI MUNTUMU	NMCP	1	<i>[Signature]</i>	9/4/25
13	Vincent Samuel Phiri	KUHEB	1	<i>[Signature]</i>	9/4/25
14	Abraham Mumba	KUHEB	1	<i>[Signature]</i>	9/4/25
15	DOROTHY DUMANDA	NMCP	1	<i>[Signature]</i>	9.4.25
16	P. H. ...	CEMAR	1	<i>[Signature]</i>	9.4.25